Project Airavat

Milestone - 1: First battery pack with integrated BMS.

* A 3S-1200mAh - 13.3Wh - 12V Li-ion battery pack.
* Meant for powering a heatsink module.
* A fail-safe 3S-40A BMS integration for preventing over-discharge or over-charging.
* The battery pack is a crucial component as it going to power a heatsink module which will contain many crucial components of the main drivetrain (an IBT-2 Motor Driver and a DC-DC Step Down Buck Converter because they tend to heat and will create heat + loss and cause inefficiencies).
* Acted as a major learning-curve for me.

→ Errors / Failures:

* Due to weak soldering joints / direct soldering on terminals of the cell, the insulation at first caused failure in circuit (mainly wires being disconnected from the terminals of the cell).
* After remaking / resoldering the whole battery pack and insulating it again, I integrated it with the BMS.
* As I was a bit overcautious of using electronics for the first time (proper industrial-electronics); I used long wires for precaution & though later on I completely sealed the wires safely (wire-management).

→ Checklist:

* B- → B_1 = 3.86V
  B- → B_2 = 7.66V
  P- → P+ = 11.67V

  } Verified with multimeter.

* A 12V Brushed DC Motor (RS-550) Stress / Load Test:

→ Checklist:

* • B- → B_1 = 3.78V
  • B- → B_2 = 7.34V
  • P- → P+ = 11.18V
  } Verified with multimeter (slight imbalance in cell-2)

* A 12V Brushed DC Motor (RS-550) Stress / Load Test:
  • Voltage Drop on Load = 11.18V → 11.01V (0.17V voltage drop)
  ∴ Very stable voltage drop even on heavy load like on a 12V RS-550 DC Motor.
  • Ran the motor for 10 minutes → No heat whatsoever detected either on pack or BMS. (0.3A current draw.)

* The over-discharge or over-charge test yet to be performed:
  •
  •

→ Conclusion:

* This battery pack build was a major milestone on this project as it was my first "perfect" battery-pack. This build taught me that EV engineering requires precision and perfection as a slight negligence can cause serious repercussions.

→ Failure: (This is a late documentation of the event so proper data is not available).

* While carrying the battery pack to the machinist's workshop who is machining the sprockets and axles for the V1 prototype of Niravat, the pack suffered a short circuit because of the bumpy roads.

* Neither the cells nor the BMS was damaged but, got a lesson

→ Analysis and Eventual Learning:

* The battery pack performed well when it was stationary, but when under bumpy road conditions (which it was meant for actually); the solder joints most probably broke off of the terminals and caused short-circuit.

* Though there were no serious repercussions but, this proves that the next iterations of 35xP configuration need to be built for brutal/heavy duty/sturdy operation and that too for a long time.

Milestone-3: A Heavy Duty 4S4P - 9600mAh - 161Wh - 16.8V Li-ion Battery Pack with 4S-20A BMS (initially) and 4S-40A BMS (final iteration). {Rechargeable}

* The main powerhouse of the Project Nirasvat.
* It is meant to power the transmitter side of the truck "Nirasvat". That means, it is powering the collective subsystem of Arduino UNO + nRF24L01 (LNA + PA) + RS-550 12V Brushed Motor + IBT-2 Driver + SG90 servo.
* It is genuinely the most stable (electrically) and reliable battery pack so far. The pack was tested for voltage sag and current draw with a "2-stage Gear Reduced, RS-550 powered electric drivetrain test rig for V2 Prototype of Nirasvat and here are the results: (My pack v/s a generic 3.7V Li-ion cell)

* This pack also utilizes precision soldering on cell terminals and is well-insulated and the wires are also well-manage
* The pack is built wi Vi-Power 3.7V-240 mAh Li-ion cells (x16).

• Voltage drop →

i). No-Load = * 0.02V (↓3.76V to 3.74V) @ 3.76V
* 0.08V (↓16.17V to 16.09V) @ 16.17V

ii). 1st - Stage = *0.01V (↓3.75V to 3.74V) @ 3.75V
(3.25:1) *0.1V (↓16.16V to 16.05V) @ 16.16V

iii). 2nd - Stage = *0.01V (↓3.75V to 3.74V) @ 3.75V
(13:1) *0.08V (↓16.16V to 16.08V) @ 16.16V

• Current Draw →

i). No-Load = * 0.15A @ 3.75V
= * 0.10A @ 16.16V

ii). 1st Stage = * 0.18A @ 3.75V (↑0.03A)
(3.25:1) = * 0.23A @ 16.16V (↑0.03A)

iii). 2nd Stage = * 0.19A @ 3.75V (↑0.04A)
(13:1) = * 0.23A @ 16.16V (↑0.03A)

* As is obvious from the data the 4S4P-Pack I created is more or less equal in performance in terms of voltage sag when compared to a generic 3.7V Li-ion cell and better in terms of current draw (the pack's current draw remained constant even under load cat 1st and 2nd stage).

→ Errors/Failures →

* A rather unrelated failure, a 4S-20A BMS was first used and didn't let the entire transmitter subsystem utilize the power from the pack as the whole TX subsystem drew more current (sometimes under load) than 20A and hence the BMS was resetting itself again and again.
* I realized that a 4S-40A BMS would be a better alternative and it turned out to be true and the performance statistics I recorded was when the 4S-40A BMS was integrated in the pack.

Milestone-4: A Custom CC/CV 0.8A-1A charger for any voltage under 19V (12.6V and 16.8V specifically)

* This charger was made with the customization of an off-the-self 19V-1.7A SMPS Power Adapter.
* A CC/CV DC-DC Buck Converter was used to adjust the voltage to 16.8V for (4S4P Pack) and 0.8A-1A through CC/CV Potentiometer and a 16Ω resistor as a load for stable customisation.

* Has a Male DC-Jack which can be inserted in the female DC-Jack of the Battery Pack to charge.
* Though not made for fast charging yet, it successfully charged the 4S4P pack from 16.00V to 16.8V w/o any heating on BMS/pack/Buck converter/adapter.

→ Errors/Failures:
* No failure detected so far; the charging is stable and no heating has been detected so far.

Milestone-5: An Arduino UNO-based transmitter subsystem utilizing RF communication, an RS-550 drive motor, an SG-90 servo and an IBT-2 Motor Controller.

* This transmitter subsystem is the brain of Niraavat V1 & V2. The central command is Arduino UNO-based.
* Detailed technical data and pin mapping:
  - The subsystem is a harmonious ecosystem of a lot of electronic modules comprising: Arduino UNO R3 + nRF24L01 (LNA+PA variant) + IBT-2 controlled - RS-550 + SG-90 servo.
  - The power flow is like this:

↳ 4S4P 16.8V Battery Pack $\xrightarrow{16.8V}$ DC-DC 5V Step Down Buck Converter $\xrightarrow{5V}$ Arduino UNO

↳ 4S4P 16.8V Battery Pack $\xrightarrow{16.8V}$ DC-DC 5V Step Down Buck Converter $\xrightarrow{5V}$ SG-90 Servo

↳ 4S4P 16.8V Battery Pack $\xrightarrow{16.8V}$ DC-DC 5V Step-Down Buck Converter $\xrightarrow{3.3V}$ nRF24L01 (LNA+PA)

↳ 4S4P 16.8V Battery Pack $\xrightarrow[(P^+, P^-)]{16.8V}$ IBT-2 + RS-550 ($P^+ \rightarrow B^+$ and $P^- \rightarrow B^-$) } BMS-Regulated direct power supply to Motor + Motor Driver

- Arduino, IBT-2 and nRF24L01 pin mapping:
(I want it known that the whole transmitter subsystem has a common GND (-ve) ground).

↳ Arduino UNO R3 → IBT-2 + RS-550

Arduino UNO R3	IBT-2
5V Power-Rail	VCC
Common-GND ground	GND
5V Power-Rail	R_EN
5V Power-Rail	L_EN
D6	RPWM
D5	LPWM
(16.8V) P+	B+
(16.8V) P-	B-
Motor's (+ve)	M+
Motor's (-ve)	M-

↳ Arduino UNO R3 → SG90 Servo (SG90-Servo $\xrightarrow{\text{Direct Power}}$ 5V Buck)

Arduino UNO R3	SG90 Servo	5V DC-DC Buck Converter
-	VCC	OUT +
Common-GND ground	GND	-
D2	Signal	-

↳ Arduino UNO R3 → nRF24L01 (LNA + PA)

(nRF24L01 was also given power directly from the 5V Buck stepped down to 3.3V. Also, a 47V-100uF capacitor was connected across VCC and GND to ensure NVH minimalization and increase stability.)

Arduino UNO R3	nRF24L01 (INA+PA)	5V DC-DC Buck Converter
Common-GND ground	GND	
-	VCC	(3.3V) OUT+
D9	CE	-
D10	CSN	-
D13	SCK	-
D11	MOSI	-
D12	MISO	-

- The important point to note is that the whole transmitter subsystem has a common GND (-ve) Power ground and nRF24L01 and SG90 Servo are powered directly from the stepped-down voltage of buck converters; which results in a more predictable and stable system overall.

→ Errors/Failures:

* It took 3 attempts to build the transmitter subsystem.
- The first attempt was a messy success and it powered up.
- The second attempt was to improve the wire improvement but leaving bare terminals (insulation wasn't done) resulted in a short circuit leading to a burnt Arduino UNO and an IBT-2.
- The third attempt finally yielded a well-managed system which booted up.

→ The instructions for operation and control/code is yet to be embedded in the transmitter subsystem. Only after embedding and running it few times could I give some reliable data.

Milestone-6: An Arduino UNO and button-based receiver module with an nRF24L01 (LNA+PA variant)

* This receiver subsystem is one of the most important subsystem in Project Nirvaat as it is responsible for sending input packets and receiving telemetry data packets and then transferring it to the Python IDE for data logging.

* Detailed technical data and pin mapping:

- The subsystem is a harmonious ecosystem of a lot of electronic modules comprising: Arduino UNO R3 + nRF24L01 (LAN+PAN variant) + 4 buttons (for controlling the motion of the truck).

- The power flow is like this:

↳ 3S3P Battery Pack $\xrightarrow{12.6V}$ 5V DC-DC Step Down Buck converter $\xrightarrow{5V}$ Arduino UNO R3

↳ 3S3P Battery Pack $\xrightarrow{12.6V}$ 5V DC-DC Step Down Buck converter $\xrightarrow{3.3V}$ nRF24L01 (LAN+PA variant)

- Pin mapping and button mapping of Arduino UNO and nRF24L01 ; and Arduino UNO and buttons:

↳ Arduino UNO R3 → Buttons:

Arduino UNO R3	Buttons
D2	Forward
D3	Back
D4	Right
D5	Left

"All buttons' GND are connected to Arduino's common GND (-ve) ground"

↳ Arduino UNO R3 → nRF24L01 (LNA+PA variant)

Arduino UNO R3	nRF24L01(LNA+PA)	5V DC-DC Buck converter
Common GND(-ve) ground	GND	–
–	VCC	OUT+(3.3V)
D9	CE	–
D10	CSN	–
D13	SCK	–
D11	MOSI	–
D12	MISO	–

- The important point to note is that the whole receiver module has a common GND(-ve) ground and nRF24L01 is powered directly from the stepped down voltage of 3.3V (from the 5V DC-DC Buck converter) and a 40V-100 uF capacitor is installed across the VCC and GND of the nRF24L01 which ensures stability and NVH-balancing.

→ Errors/Failures:

* Learning from the mistakes made while making the transmitter subsystem resulted in successful making and testing up of the receiver subsystem in single attempt.

→ The code is yet to be embedded in the receiver subsystem. Only after embedding and running it few times could I give some reliable data.

#Milestone-6: A Heavy Duty 3S3P - 7200mAh - 91 Wh - 12.6V Li-ion Rechargeable Battery Pack with 3S-40A BMS - protection.

* A backup powerhouse for Project Niraavat.
* This battery-pack is the statement of "rapid prototyping" at a small scale. I built this battery pack from scratch; with all the soldering, wiring, off-the-shelf 3S-40A BMS integration and multimeter verification; within 30 minutes though, the insulation and wire management took 20 minutes later on; so overall it took less than an hour to build the pack from scratch.
* This pack also utilizes precision soldering on cell terminals and is well-insulated and the wires are also well-managed.
* The pack is built with Vi-Power 3.7V - 2400mAh Li-ion cells (x9).

→ Voltage sag, current Draw and other performance parameters are yet to be recorded.

→ Errors / Failures:

* No errors/failures detected so far.

→ Checklist:

- B- → B_1 = 4.02V
- B+ → B_2 = 8.03V
- B- → B+ = 12.04V
- P- → P+ = 11.75V (Internal resistance of protective MOSFETs resulted in ca. difference of 0.29V from B- → B+ and P- → P+.)

#Milestone-7: A 3S3P - 3600mAh - 45A/h - 12.6V Li-ion Rechargeable Backup Battery Pack with 3S-40A BMS-protection.

* A generic testing/backup battery pack for testing subsystems of Project Niraavat.
* This pack also utilizes precision soldering on cell terminals and is well-insulated and the wires are also well managed.
* The pack is built with generic Hong-Li 3.7V - 1200mAh Li-ion cells (x9)

→ Voltage Sag, Current Draw and other Performance Parameters never tested (pack wasn't meant to be a high-performance unit; just a testing/backup battery pack).

→ Errors/Failures: No errors/failures detected so far.

→ Checklist:

- B- → B_1 = 3.8V
- B- → B_2 = 7.7V
- B- → B_+ = 11.75V
- P- → P+ = 11.5V

Milestone-8: Proof-of-concept of the 1-stage Reduction of the "2-stage Jackshaft-based sprocket-chain reduction - RS-550-based Electric Powertrain" dey a 14:12 (1.16) ratio 14-teeth and 12-teeth sprocket and chain based electric drivetrain.

* Though the 1st-stage Reduction was supposed to be a 60-teeth sprocket (mounted on the jackshaft) connected directly to the 12-teeth pinion sprocket mounted on an RS-550's shaft; due to certain circumstances, only the 14-teeth sprocket and 12-teeth sprocket with their axles were delivered after machining. So, I proceeded to make a proof-of-concept 14:12 (1.16) ratio sprocket-chain drivetrain.

* The shaft was obviously inserted in/mounted on a 12mm bore bearing and the 14-teeth sprocket mounted on it (the jackshaft) and tightened temporarily with hot glue though, the sprocket does utilizes grub screw configuration, the circumstances didn't allow me to utilize (I was in a hurry to validate the concept) and mounted the motor with 12-teeth pinion sprocket and connected both the sprockets with a chain and the drivetrain surprisingly ran despite many inconsistent approaches and inefficient mountings.

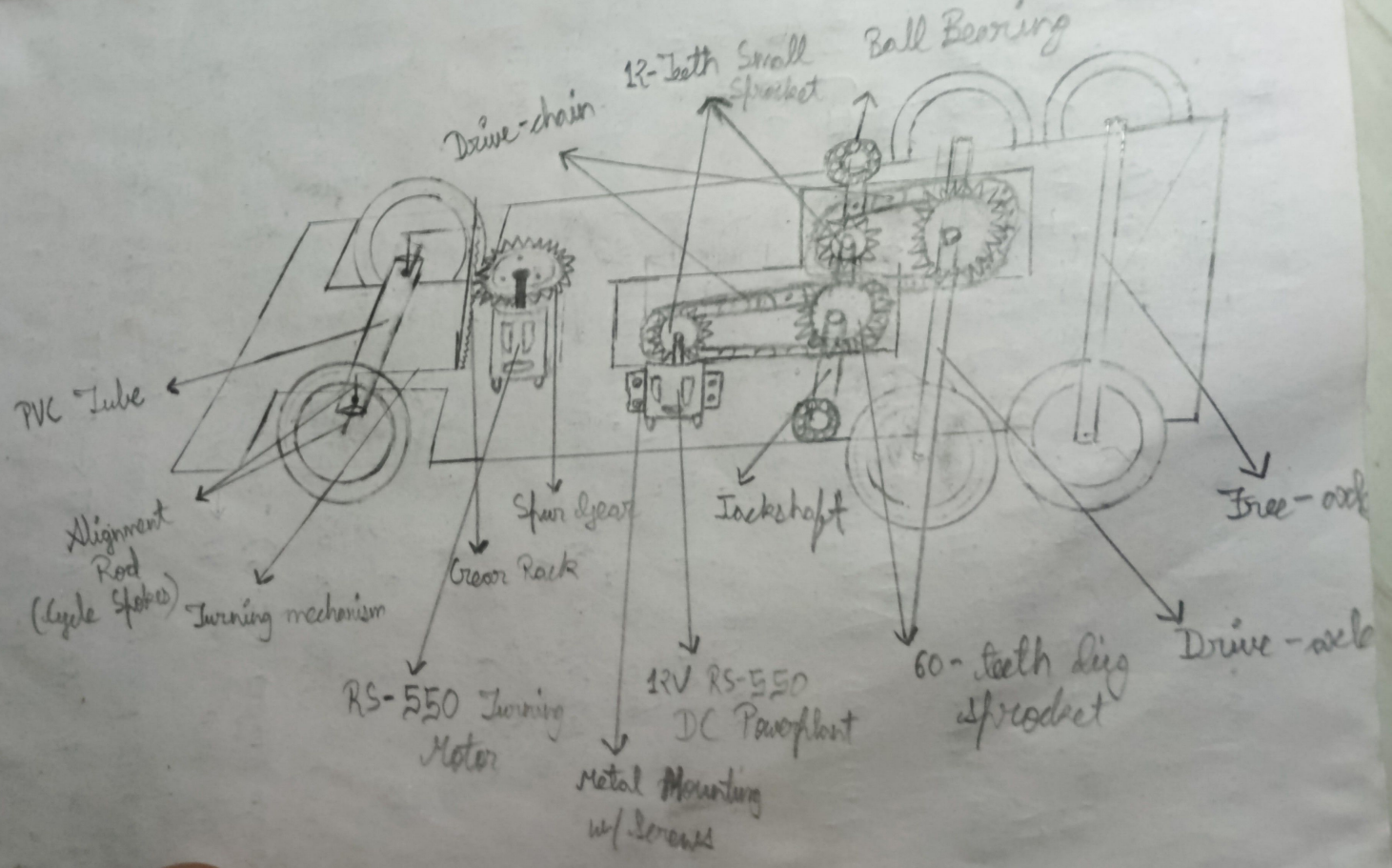
→ Checklist:

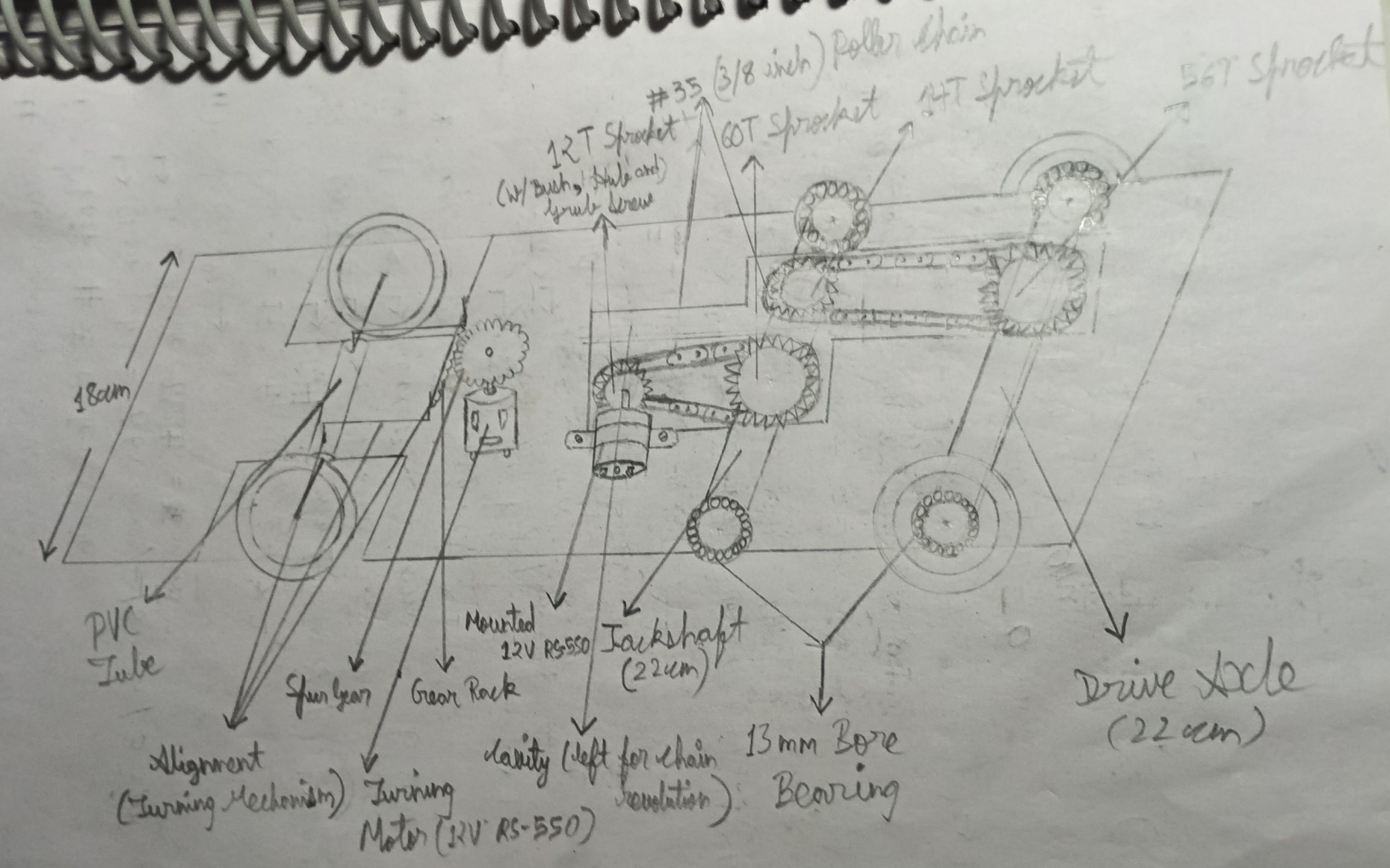
- The concept of 2-stage reduction through jackshaft chain sprocket-chain system was validated as the chain didn't derail and the shaft was moving nice and smooth (which indicates proper chain and shaft alignment and proper sprocket geometry).
- Machined sprocket's geometry is validated.

→ Inefficiencies and points to focus on:

- The shaft was wobbling left and right on 100% throttle, which introduced chain and shaft-axis misalignment.
- The misalignment caused NVH (medium-high) in motor.
- Using a recycled timing chain from a scrap motorcycle without any cutting and riveting caused utilization of unnecessary length of the chain.
- Because of excessive chain length; chain slack, chain slap and other inefficiencies related to the chain took place.
- Proper industry-standard fastening and mounting should be utilized later on during the actual prototyping, not any temporary measures.

Drivetrain Design




#35 (3/8 inch) Roller chain
12T Sprocket
(w/ Bush, Hub and Grub Screw
60T Sprocket
14T Sprocket
56T Sprocket
18cm
PVC Tube
Spur Gear
Gear Rack
Mounted 12V RS-550
Jackshaft (22cm)
Drive Axle (22cm)
Alignment (Turning Mechanism)
Turning Motor (12V RS-550)
Clarity (left for chain revolution)
13mm Bore Bearing

Drivetrain Design - II

Specifications of Drivetrain:

i). Chain: #35 Roller Chain
- ↳ Chain pitch = 7.62 mm / 318 inch / 3.
- ↳ Inner/Roller Width = 8 mm

ii). Sprockets →

* Stage-1 = 12 Teeth Sprocket × 2
  - ↳ Width = 5 mm (Tooth gap) ↳ Bore = 12 mm
  - 1.42 ← ↳ Plate Thickness = 4.5 mm
  - 11 ← ↳ Diameter = 35 mm

  { Sprocket with a hub and grind to shaft and grub screw. }

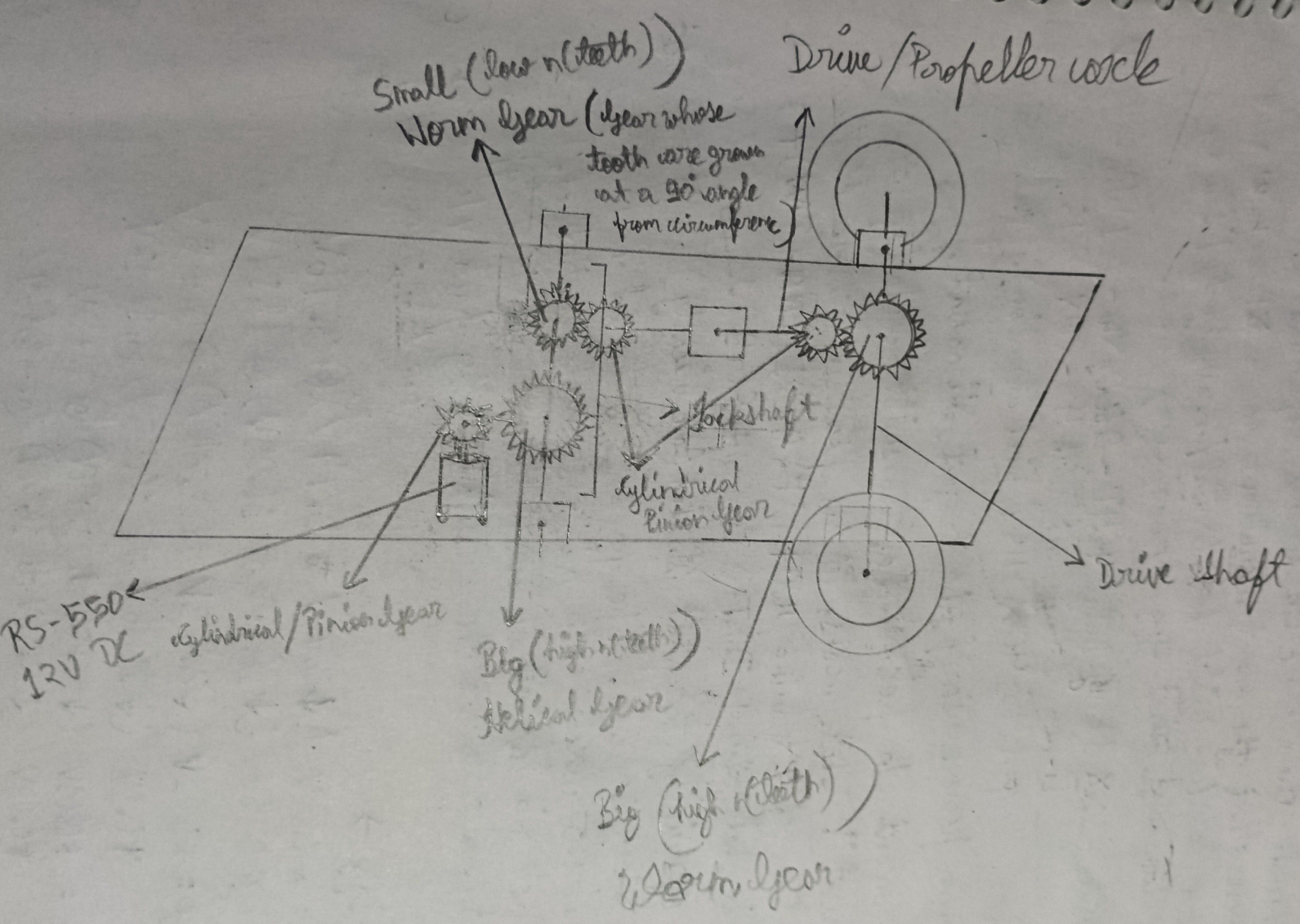
* Stage-2 = 60 teeth Sprocket × 1
  - ↳ Width/Tooth gap = 5 mm ↳ Bore = 12 mm
  - 2 → ↳ Plate Thickness = 6 mm
  - 47.24 ↳ Diameter = 15 cm

  = 14 Teeth Sprocket × 1
  - ↳ Width = 5 mm (Tooth gap) ↳ Bore = 12mm
  - 1.42 ↳ Plate Thickness = 4.5 mm
  - 11 ↳ Diameter = 35 mm

* Drive Stage = 56 Teeth Sprocket × 1
  - ↳ Width = 5 mm (Tooth gap) ↳ Bore = 12mm
  - 2 → ↳ Plate Thickness = 6.0 mm
  - 37.8 → ↳ Diameter = 12

iii). Jackshaft/Axle = 12 mm aluminium × 2 (22 cm each)
4.1

iv). Bearing = 12mm bore × 4


Small (low n(teeth))
Worm Gear (Gear whose teeth are grown at a 90° angle from circumference)
Drive/Propeller axle
Lockshaft
Cylindrical Pinion Gear
RS-550 12V DC
Cylindrical/Pinion Gear
Big (high n(teeth))
Helical Gear
Drive Shaft
Big (high n(teeth))
Worm Gear